

Name: _____ Suvit Kumar _____ Date: ____3/8/2026____ Class: _____
_____ Engineering _____

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The design needs to be stable. This is the highest priority tied with the dimension requirements. This will lead to a proper solution as this is used by kids, and kids are not known for being careful. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #2 (HIGH Priority)

Any part of the design that the kids have to access has to be 18 inches or shorter, 10 inches in length, 14 inches in width. This is also the highest priority tied with being stable as this needs to be usable. This will lead to a valid solution as if it did not follow the length and width requirements, it would not fit on the desk, and if it did not follow the height requirement, it would be unusable for the shorter kids. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #3 (HIGH Priority)

The design needs to be durable. This is the second highest priority since kids are using this, so it should not break. This will lead to a valid solution as without durability, the kids will soon break it causing it to be unusable for kids, and this is a machine for kids and would fail if it were unusable or have to be replaced often. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #4 (HIGH Priority)

The design needs to be simple for the kids to use. This is the third highest priority since kids are using this, so this needs to be simple. This will lead to a valid solution as without being simple to use, the kids will not use it thus it will fail. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #5 (HIGH Priority)

The design needs to be low-maintenance for the librarian. This is the fourth highest priority since the librarian also has to keep the questions going and all. This will lead to a valid solution as without this, the solution cannot be kept for a long period of time as eventually it would become too much work. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #6 (LOW Priority)

The design needs to hold ~300 pom poms. This is a low priority at only fifth highest priority as it is not that important compared to the rest. This will still lead to a valid solution as 300 pom poms is at the very least the minimum daily amount of kids, and the librarian stated the maximum frequency of being replaced is daily. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

Design Requirement #7 (LOW Priority)

The design needs to clearly show who's winning. This is the lowest priority and it is not that important. This will lead to a valid solution as the librarian stated it would be nice if the design could clearly show who is winning. When this design requirement was presented, there was no negative feedback about this design requirement, signaling stakeholder validation.

